MTH309 PRACTICE PROBLEMS WEEK 10

Question 1. Find examples of random variables $X, X_1, X_2, \cdots$ on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that the indicated properties hold.

(a) $X_n \xrightarrow[n\to\infty]{\text{in probability}} X$, but $X_n \nrightarrow X$ in $\mathcal{L}^1(\mathbb{P})$.

(b) $X_n \xrightarrow[n\to\infty]{\text{in probability}} X$, but $X_n \nrightarrow X$ a.s.

Question 2. Let $X, X_1, X_2, \cdots$ be random variables on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Prove the following equivalences.

(a) $X_n \xrightarrow[n\to\infty]{\text{in probability}} X \iff \mathbb{E}\frac{|X_n - X|}{1+|X_n-X|} \xrightarrow{n\to\infty} 0$.

(b) $X_n \xrightarrow[n\to\infty]{\text{in probability}} X \iff \mathbb{E}\left(|X_n - X| \wedge 1\right) \xrightarrow{n\to\infty} 0$.

(c) $X_n \xrightarrow[n\to\infty]{\text{in probability}} X \iff \inf\{\epsilon > 0 : \mathbb{P}(|X_n - X| > \epsilon) < \epsilon\} \xrightarrow{n\to\infty} 0$.

Note: One can now formulate metrics on the set of random variables, the convergence in which are equivalent to the convergence in probability. These metrics are sometimes referred to as the Ky Fan metrics.

Question 3. Let $X, X_1, X_2, \cdots$ be random variables on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Prove the following equivalences.

(a) $X_n \xrightarrow[n\to\infty]{\text{a.s.}} X$ if and only if for all $\epsilon > 0$,

$$\lim_{m\to\infty} \mathbb{P}(|X_n - X| \le \epsilon, \forall n \ge m) = 1.$$

(b) $X_n \xrightarrow[n\to\infty]{\text{a.s.}} X$ if and only if for all $\epsilon > 0$,

$$\lim_{m\to\infty} \mathbb{P}(|X_n - X_k| > \epsilon, \text{ for some } n, k \ge m) = 0.$$

(c) $X_n \xrightarrow[n\to\infty]{\text{in probability}} X$ if and only if for all $\epsilon > 0$,

$$\lim_{n,k\to\infty} \mathbb{P}(|X_n - X_k| > \epsilon) = 0.$$